

SMART LIBRARY REQUEST WORKFLOW IN SERVICENOW

Abstract

The Smart Library Request Workflow in ServiceNow is designed to modernize the traditional library management process by introducing automation, digital workflows, and centralized data management. In many educational institutions, the process of requesting, approving, and issuing books is still performed manually. This leads to inefficiencies, delays, and difficulties in tracking book availability.

This project proposes a workflow-based library system implemented on the ServiceNow platform. The system enables students to request books through an online portal, while librarians manage approvals, maintain book inventories, and track issued and returned books through automated workflows. The platform ensures transparency, real-time updates, and improved efficiency in library operations.

By implementing automation and role-based access control, the system minimizes manual work and enhances the overall user experience for both students and library administrators.

1. Introduction

Libraries play an important role in educational institutions by providing access to knowledge resources such as books, journals, and digital materials. However, many libraries still rely on manual processes for managing book requests, issue records, and inventory management.

Manual systems often lead to problems such as misplaced records, delays in issuing books, and lack of transparency regarding book availability. Students may need to physically visit the library to check whether a book is available. Similarly, librarians spend significant time maintaining registers and tracking issued books.

The Smart Library Request Workflow project aims to solve these issues by introducing a digital workflow-based system using the ServiceNow platform. ServiceNow provides tools to create applications with automated workflows, forms, and approval processes. By using these features, the library management process can be streamlined and made more efficient.

The system allows students to submit book requests online, after which librarians review and approve them. Once approved, the system updates the inventory and records the transaction automatically.

2. Existing System

In many colleges and institutions, libraries still operate using manual or semi-digital systems. These systems often involve maintaining registers or spreadsheets to track book issues and returns.

Problems in the existing system include:

- Lack of real-time information about book availability.
 - Time-consuming manual record maintenance.
 - Difficulty in tracking overdue books.
 - Lack of automated notifications.
 - Limited reporting and analytics.
- Because of these limitations, both students and librarians face inefficiencies when accessing or managing library resources.

3. Proposed System

The proposed system introduces a Smart Library Request Workflow built using the ServiceNow platform. The system digitizes the entire process of requesting, approving, issuing, and returning books.

Key features of the proposed system include:

- Online book request submission by students.
 - Automated approval workflow for librarians.
 - Centralized database for managing books and users.
 - Real-time tracking of issued and returned books.
 - Automated notifications for request status.
 - Dashboard and reporting features for library administrators.
- By implementing this system, libraries can significantly improve efficiency, reduce paperwork, and provide better services to students.

4. Objectives of the Project

The main objectives of this project are:

- To develop a digital library request system using ServiceNow.
- To automate the book request and approval workflow.

Empathy Map Canvas

The Empathy Map Canvas is used to understand the needs, expectations, and challenges of users interacting with the library management system. In this project, the main users are students and librarians. By analysing their perspectives, the system design can be improved to provide a better and more efficient user experience.

Students are the primary users who request books from the library. They expect a simple and fast way to search for books and submit borrow requests. Students often think about completing their academic work and need quick access to study materials. They may feel frustrated when they cannot easily determine whether a book is available or when the approval process takes too long. Therefore, students require a system that clearly shows book availability, allows easy request submission, and provides real-time updates on request status.

Librarians are responsible for managing the library resources. They handle tasks such as maintaining the book catalog, approving borrow requests, and ensuring books are returned properly. Librarians think about maintaining accurate records and managing multiple requests efficiently. Without a digital system, manual processes can create confusion, delays, and errors in record keeping.

Through the empathy map analysis, the project identifies what users think, say, feel, and do when interacting with the system. Students want faster service and transparency, while librarians need efficient tools to manage the library catalog and requests. Understanding these user behaviours helps in designing features such as automated workflows, role-based access control, and real-time status updates.

By applying the Empathy Map Canvas, the project ensures that the system focuses on solving real user problems. The Smart Library Request Workflow therefore provides a user-friendly interface and efficient process that benefits both students and librarians.

Brainstorming – Idea Generation – Prioritization Template

Brainstorming is a creative process used to generate multiple ideas that can help solve a particular problem. In this project, brainstorming was conducted to identify possible solutions for improving the traditional library management system. The main objective was to find ways to automate book borrowing, simplify request handling, and improve the efficiency of library operations.

During the brainstorming phase, several ideas were discussed. One idea was to create an online platform where students can search for books and submit borrow requests digitally. Another idea involved automating the approval process using workflow automation so that librarians can review requests quickly. Additional ideas included implementing role-based access control, generating reports on library usage, and sending automated notifications when requests are approved or rejected.

After generating different ideas, the next step was idea evaluation and prioritization. Not every idea can be implemented immediately, so the team evaluated each idea based on feasibility, impact, and implementation complexity. Ideas that offered the highest benefit to users and were technically possible within the ServiceNow platform were given priority.

Based on this prioritization process, the project selected key features such as automated borrow request workflows, role-based permissions, reference qualifiers to ensure only available books can be requested, and reporting features like “Most Borrowed Books.” These features directly address the major problems faced in manual library systems.

The brainstorming and prioritization process helped transform creative ideas into a practical system design. It ensured that the final solution focuses on the most valuable features that improve library management efficiency and user satisfaction.

Define Problem Statements Template

The Define Problem Statements Template helps clearly identify the problems that need to be solved by the project. In the case of the Smart Library Request Workflow system, the main problem arises from the traditional manual management of library operations.

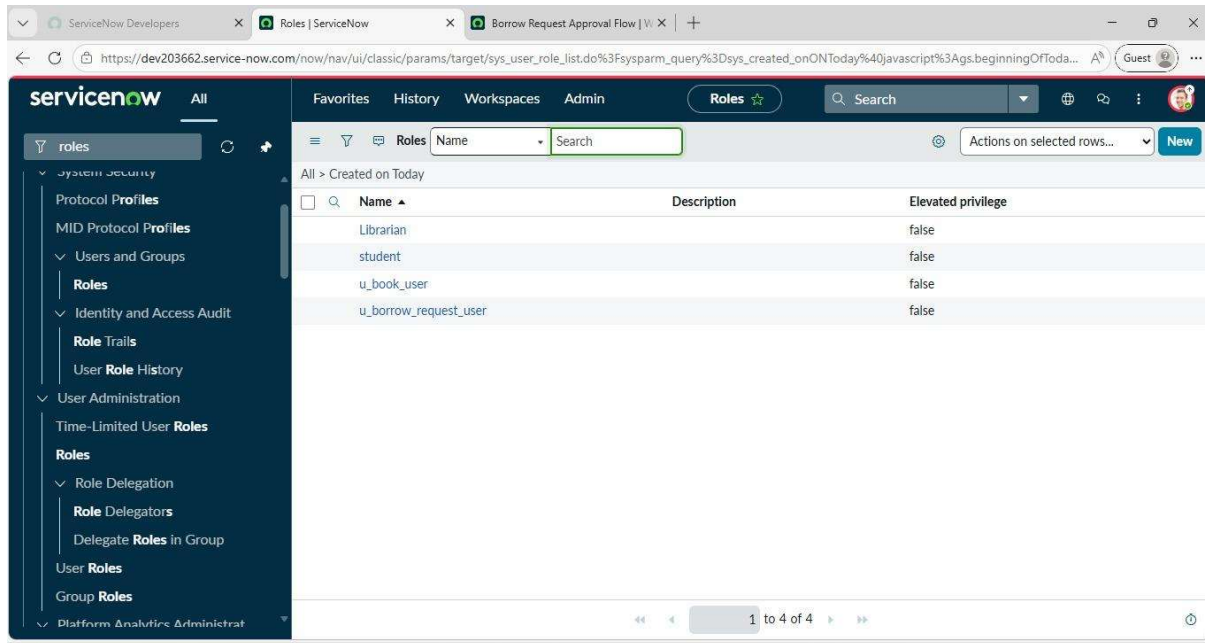
In many institutions, library records are maintained manually or through limited digital systems. This approach often results in inefficiencies such as difficulty in tracking book availability, delays in processing borrow requests, and lack of transparency in the borrowing process. Students may struggle to find out whether a book is available or already issued, which leads to wasted time and repeated inquiries.

Librarians also face challenges in managing multiple requests and maintaining accurate records. Manual record keeping increases the risk of errors and makes it difficult to monitor borrowing patterns or generate useful reports. Additionally, without automated workflows, approval processes become slow and inefficient.

The defined problem statement for this project is the need for a digital library management system that automates the process of book borrowing and management. The system should allow students to easily request books, while librarians manage approvals and update book status. It should also provide real-time visibility of book availability and generate reports for better decision-making.

To solve this problem, the project implements a role-based system on the ServiceNow platform with automated workflows, secure access control, and reporting features. This solution improves efficiency, reduces manual work, and enhances the overall experience for both students and librarians.

ACTIVITY 1 – ROLE CREATION



Role Creation is the process of defining user roles and their permissions in a system. It determines what actions a user can perform and what data they can access. This concept is commonly used in platforms like ServiceNow, databases, and other enterprise systems.

1. What is Role Creation?

Role creation means creating a specific role for users based on their responsibilities. Each role has certain permissions that allow users to perform particular tasks in the system.

For example in a Smart Library System:

Student Role – Can request or borrow books.

Librarian Role – Can approve requests and manage books.

Admin Role – Can manage users and system settings.

2. Purpose of Role Creation

Role creation is used to:

Control user access to the system.

Improve security by limiting permissions.

Ensure users perform only authorized actions.

Organize system management efficiently.

3. Steps in Role Creation (Example in ServiceNow)

Log in as an Administrator.

Navigate to User Administration → Roles.

Click New to create a role.

Enter the Role Name and description.

Assign the role to users or groups.

Save the role.

ACTIVITY 2 - TABLE CREATION

The screenshot shows the ServiceNow interface for creating a new table. The left sidebar contains navigation options like 'Data Sources', 'Protected Tables', 'System Archiving', 'Archive Tables', 'System Clone', 'Clone Definition', 'Exclude Tables', and 'System Definition'. The main area is titled 'Table - Book' and includes a search bar and buttons for 'Delete', 'Update', and 'Delete All Records'. Below the title, there is a description: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'. The 'Columns' tab is selected, showing a list of 'Table Columns' with a search bar and a 'New' button. The 'Dictionary Entries' table lists columns with their labels, types, references, max lengths, default values, and display settings.

Column label	Type	Reference	Max length	Default value	Display
Title	String	(empty)	40		true
Status	Choice	(empty)	40		false
Created by	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false

Table Creation is the process of creating a database table to store and organize data in a structured format. A table contains rows (records) and columns (fields) where information is stored.

1. What is Table Creation?

Table creation means designing a table with specific fields to store related data. Each column represents a type of information, and each row represents a single record.

For example in a Smart Library System:

A table can store book request details such as student name, book name, request date, and status.

2. Purpose of Table Creation

Table creation is used to:

Store system data in an organized way

Manage records efficiently

Retrieve and update information easily

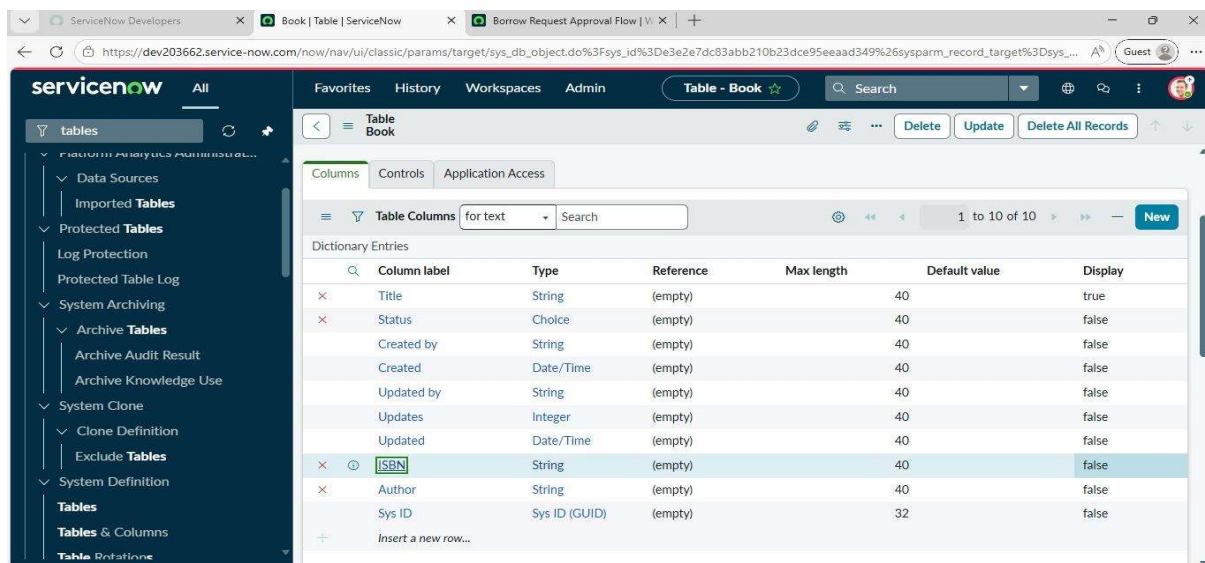
Support system workflows and automation

3. Example Table (Library Request Table)

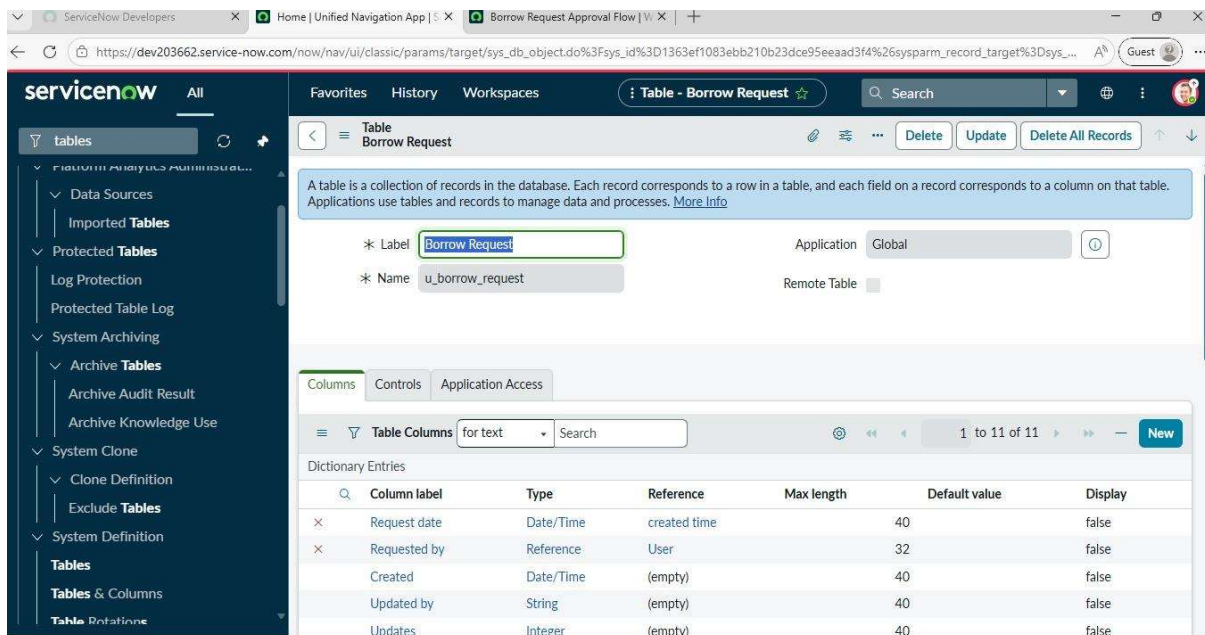
Request ID	Student Name	Book Name	Request Date	Status
001	Arun	Data Science	10-02-2026	Pending
002	Priya	Python	11-02-2026	Approved
003	Karthik	Machine Learning	12-02-2026	Rejected

4. Steps for Table Creation in ServiceNow

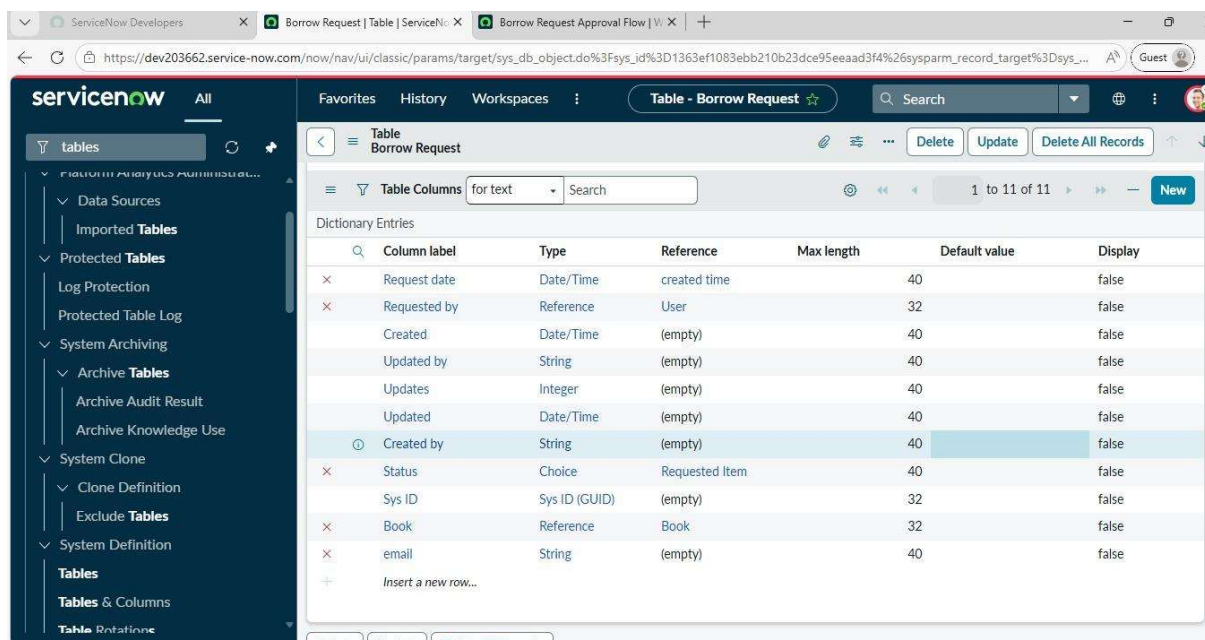
1. Log in as Administrator
2. Navigate to System Definition → Tables
3. Click New Table
4. Enter the table name and label
5. Add required fields (columns) such as name, date, status
6. Save the table



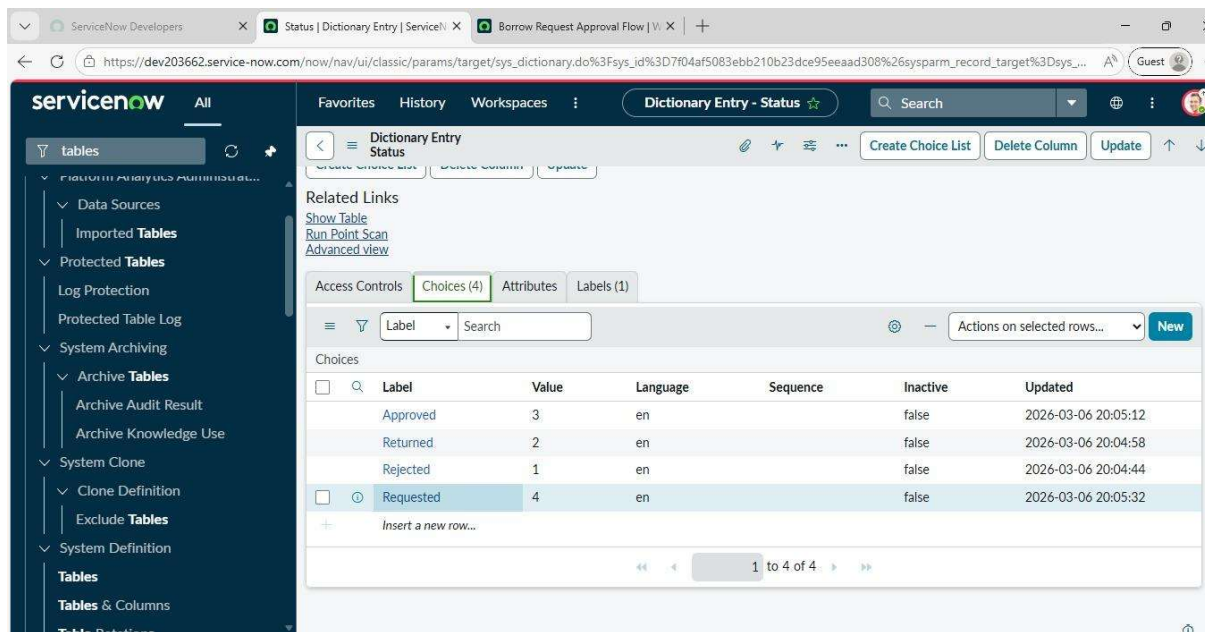
- This image displays the columns created for the Book table.
- Important fields such as Title, Author, and Status are configured.
- Each column stores specific information about a book.
- These fields help maintain structured data in the database.



- This image shows the creation of the Borrow Request table.
- This table is used to record requests made by users to borrow books.
- It stores information such as the requester and request date.
- This helps track borrowing activities in the system.



- This image displays the fields added to the Borrow Request table.
- Fields like requested date, requested by, and status are configured.
- These fields help monitor the progress of each request.
- It makes the borrow request process organized.



- This image shows the configuration of the Status choice list.
- Options such as Requested, Approved, Rejected, and Returned are added.
- These values represent the different stages of a borrow request.
- Choice lists help maintain consistent data.

ACTIVITY 3 - FLOW REQUEST

Flow Report

A Flow Report shows how a workflow process runs inside the system. It displays the steps completed in a workflow, the status of requests, and the actions performed by users or the system.

Purpose of Flow Report

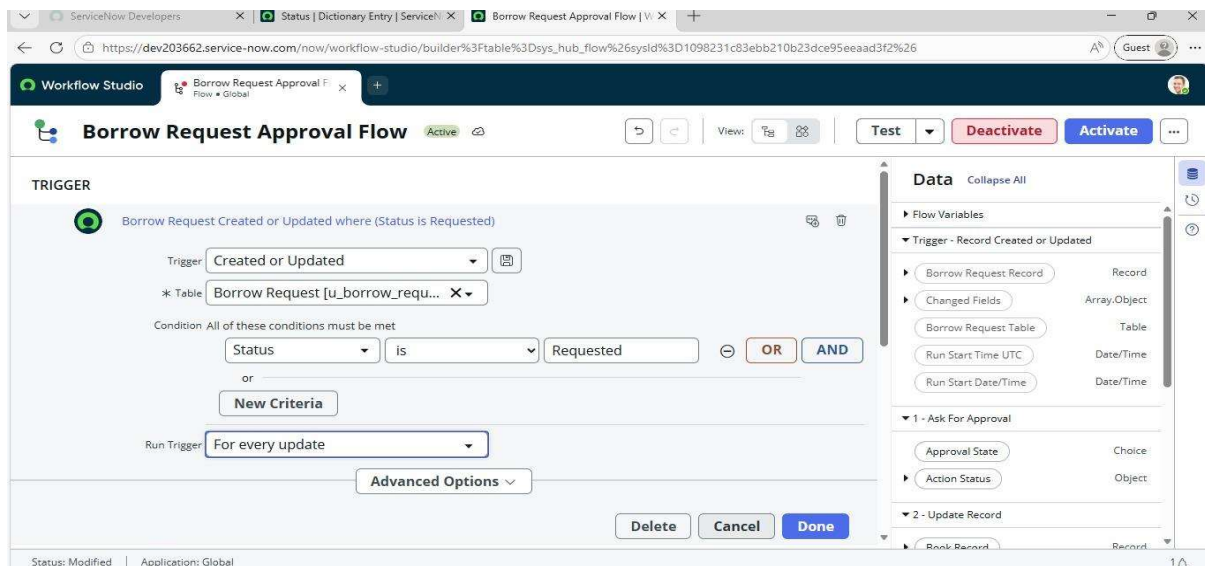
- Monitor the progress of workflows.
- Track request approvals and rejections.
- Identify errors or delays in the process.
- Provide a clear overview of system activities.

Example in Smart Library System

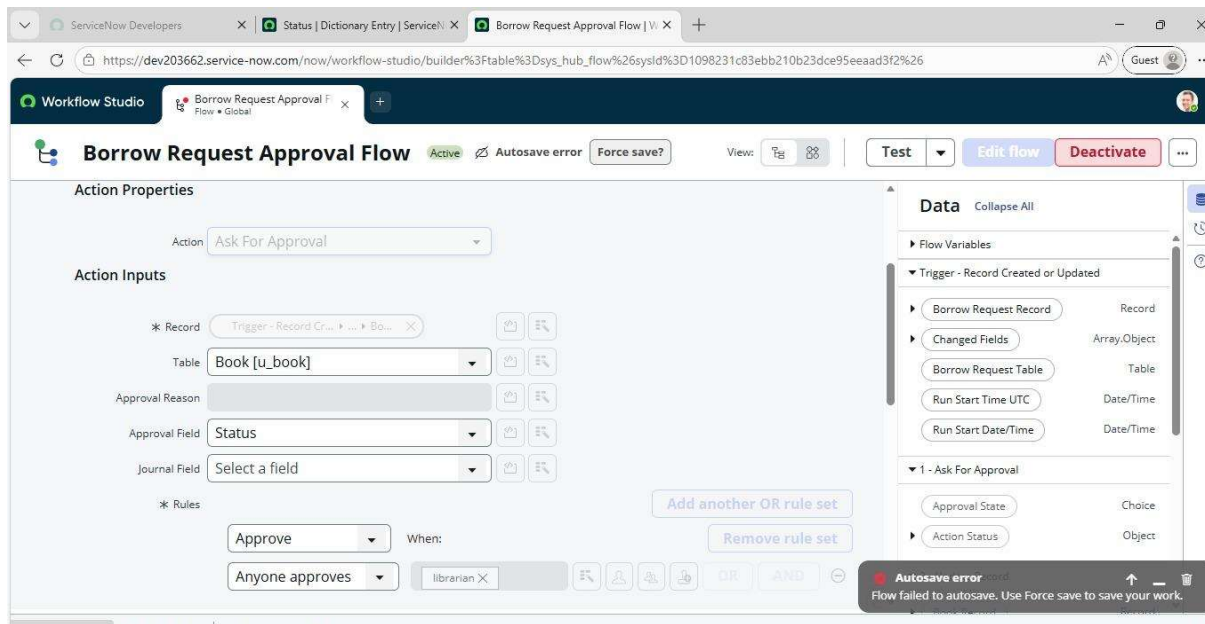
In the Smart Library Request Workflow:

1. A student submits a book request.
2. The request moves to the librarian for approval.
3. The librarian approves or rejects the request.
4. The system updates the book status.

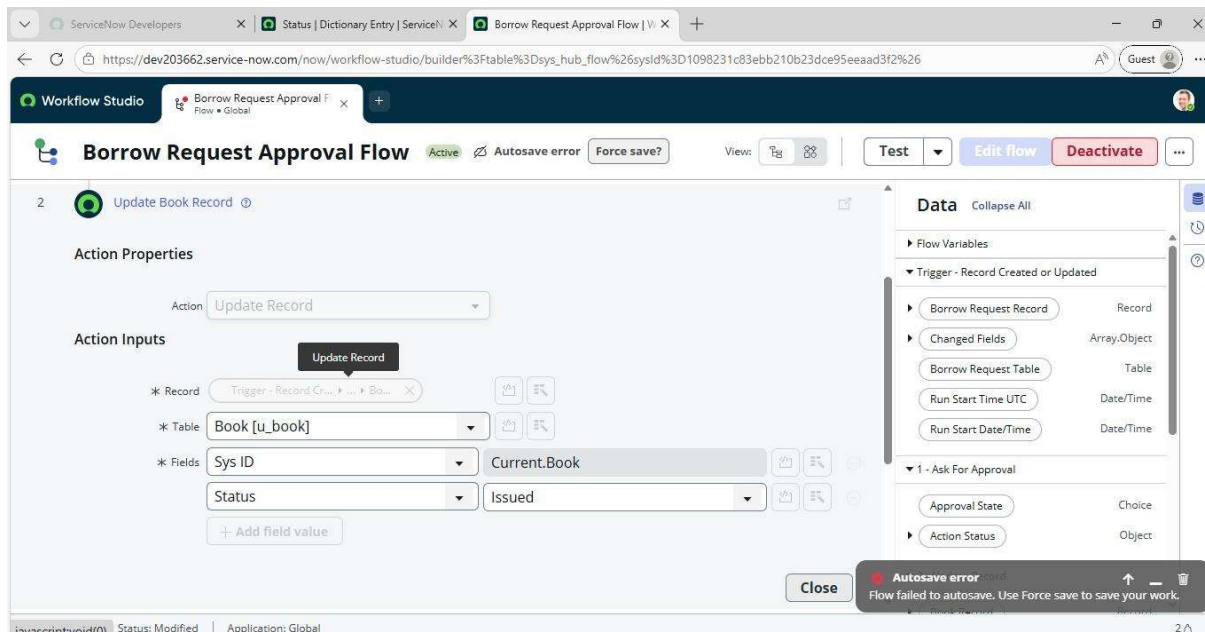
The Flow Report shows all these steps with their status (pending, approved, completed).



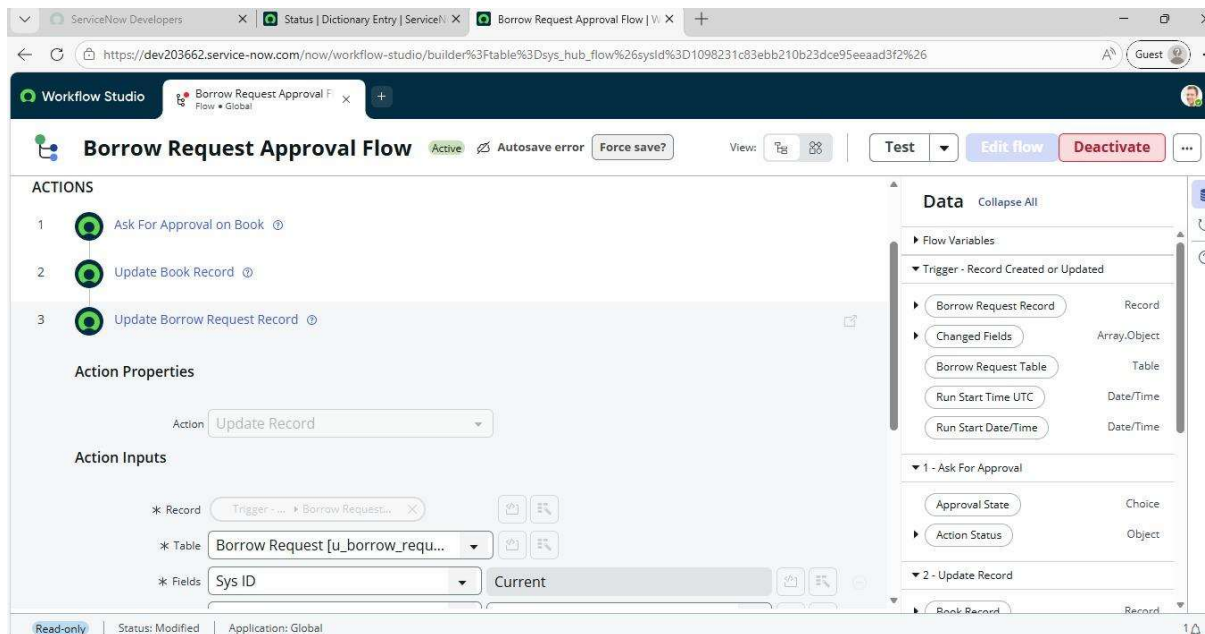
- This image shows the workflow or flow designer used for the Borrow Request approval process.
- The workflow is created to manage how borrow requests are processed.
- It includes steps for approval and status updates.
- Automation helps reduce manual work.



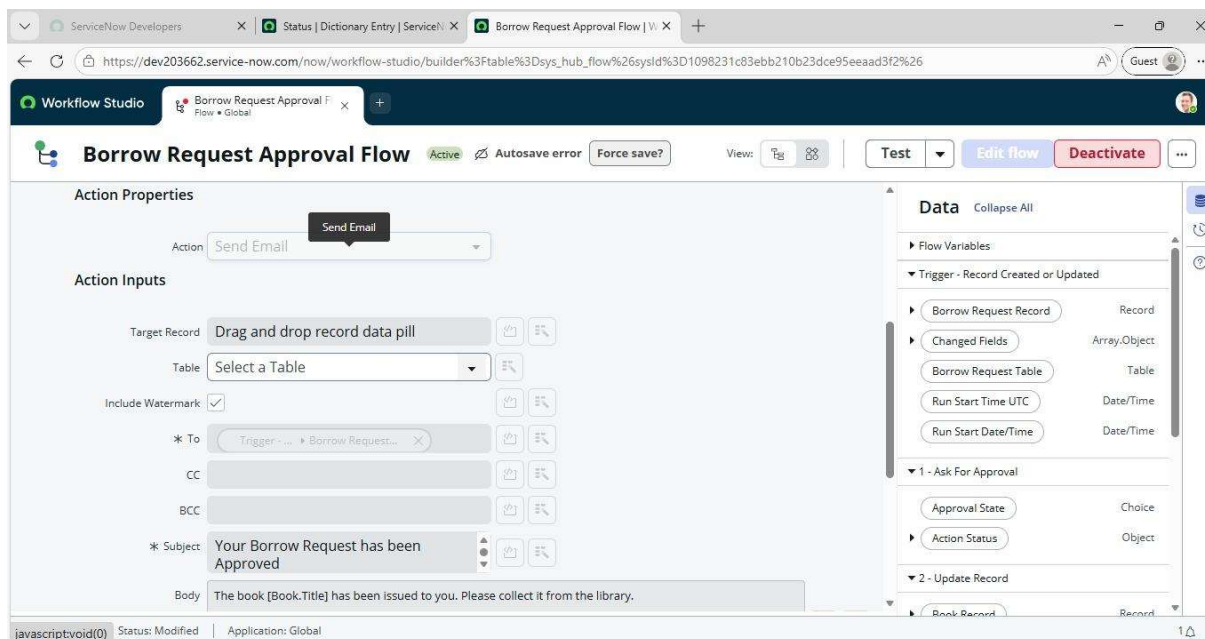
- This image displays the trigger configuration for the workflow.
- The flow starts when a Borrow Request record is created or updated.
- It specifically triggers when the request status is set to Requested.
- This ensures the approval process begins automatically.



- This image shows an action that updates the Book record.
- When a request is approved, the book status is changed to Issued.
- This indicates that the book is currently borrowed by a user.
- The workflow automatically updates this information.



- This image displays multiple actions inside the workflow.
- Actions such as approval request and record update are included.
- These actions are executed in sequence.
- This ensures that each borrow request follows the proper process.



- This image shows the email notification configuration.
- When a borrow request is approved, an email is sent to the user.
- The notification informs the user about the approval status.
- This improves communication within the system.
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ACTIVITY 4 - UI POLICY

UI Policy (User Interface Policy) is a feature used to control the behavior of fields on a form automatically based on certain conditions. It is commonly used in platforms like ServiceNow to make forms dynamic and user-friendly.

1. What is UI Policy?

A UI Policy is a rule that changes the visibility, mandatory status, or read-only property of form fields when specific conditions are met. It helps guide users to enter correct information.

2. Purpose of UI Policy

- Make forms easier to use
- Ensure required information is filled
- Hide unnecessary fields
- Improve data accuracy

3. Example in Smart Library System

- In a Smart Library Request Workflow:
- If Request Type = Borrow Book, the Book Name field becomes mandatory.
- If Request Status = Approved, the Return Date field becomes visible.
- This ensures the user fills only the relevant details.

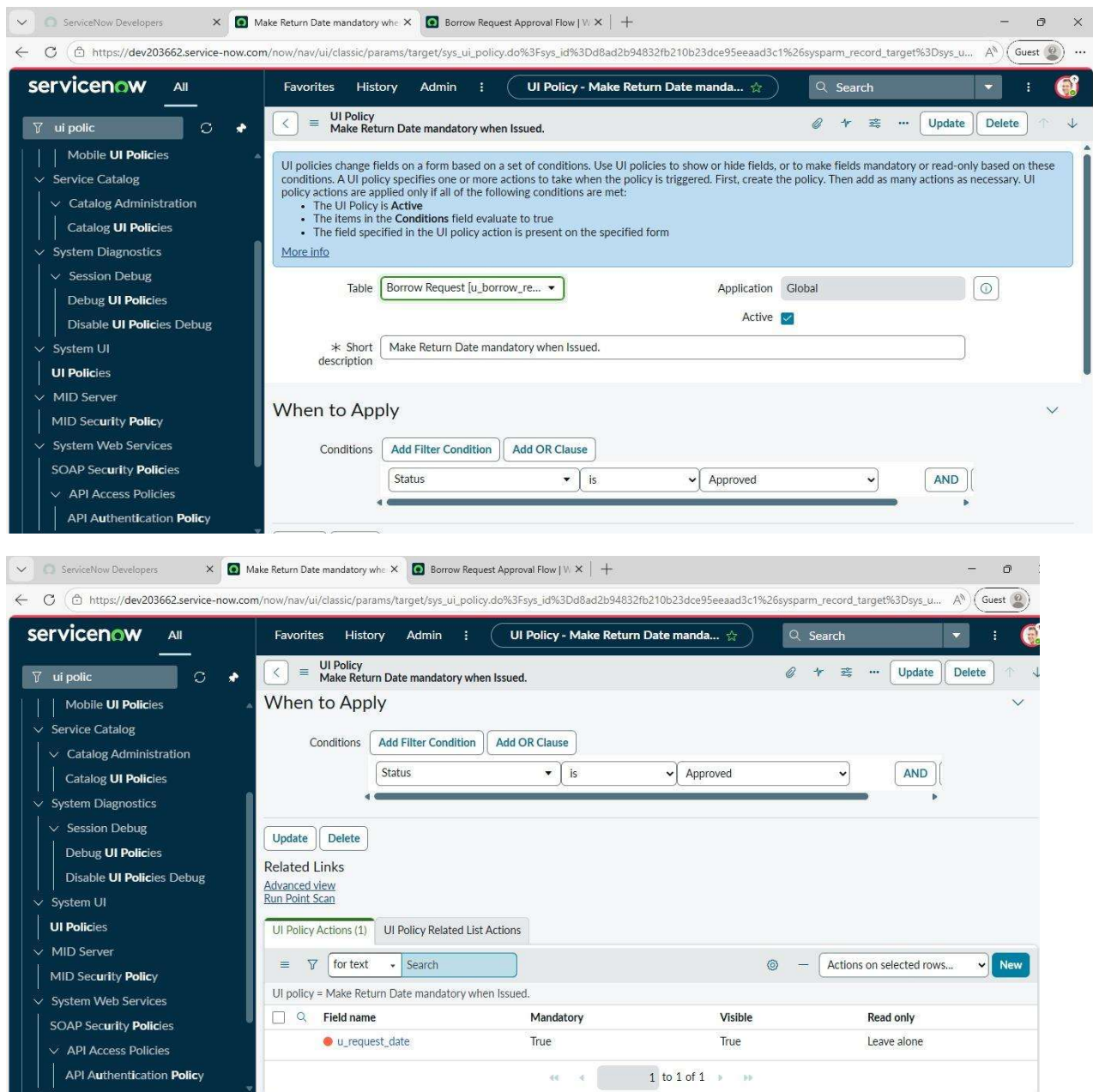
4. Steps to Create UI Policy

1. Go to System UI → UI Policies
2. Click New
3. Select the table where the policy will apply
4. Define the condition
5. Set field actions (mandatory, visible, read-only)
6. Save the policy

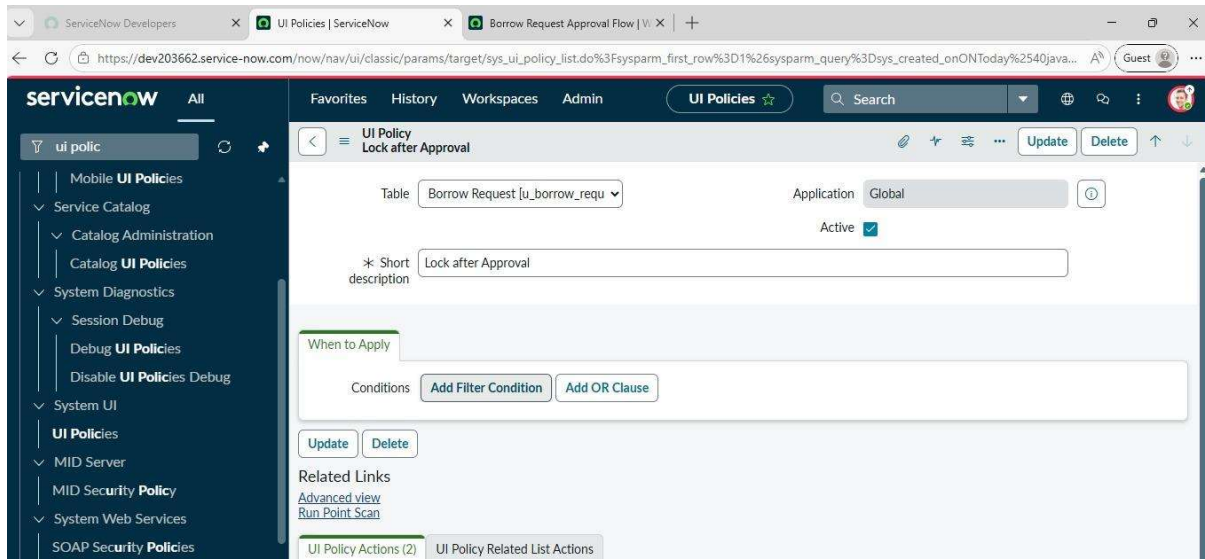
5. Benefit

5. Benefits

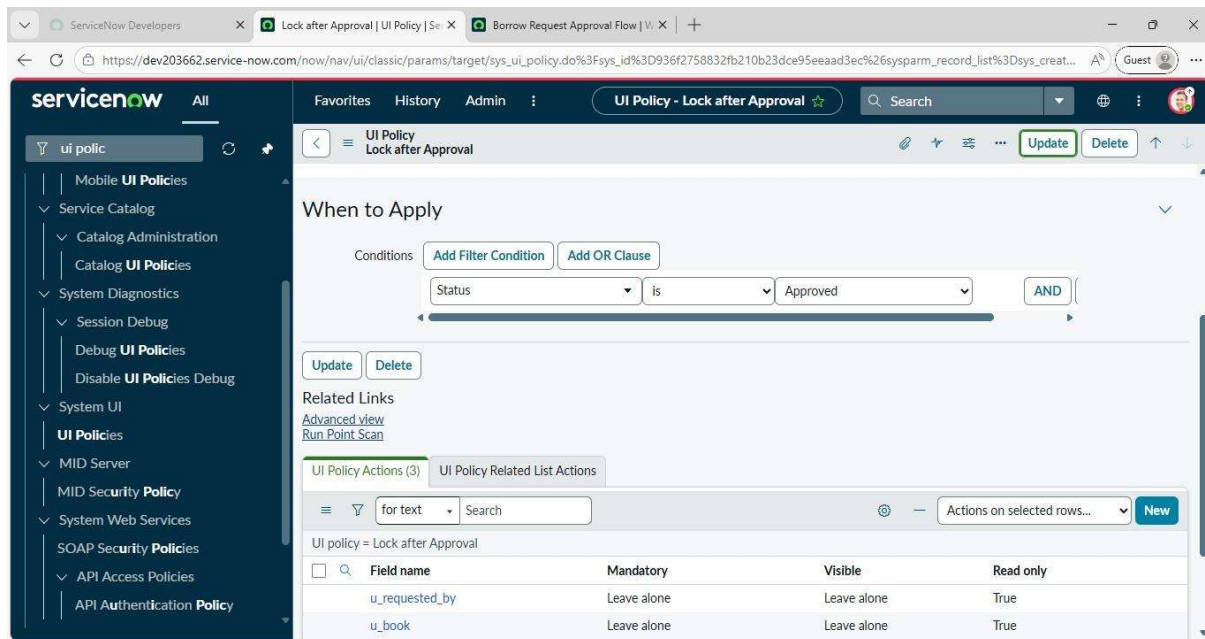
- Improves form usability
- Reduces user errors
- Ensures correct data entry
- Automates form behavior



- This image shows the creation of a UI Policy in ServiceNow.
- The policy ensures that the return date field becomes mandatory when the book is issued.
- It helps enforce required data entry.
- This ensures that the return date is always recorded.



- This image displays the conditions set for the UI Policy.
- The policy is triggered when the request status changes to Approved.
- Once triggered, the return date field becomes mandatory.
- This improves data accuracy.



- This image shows another UI Policy called Lock After Approval.
- This policy prevents users from editing certain fields after approval.
- It protects important data from being changed.
- This helps maintain record integrity.

Activity 5 - ACCESS CONTROL UNIT

Access Control (ACL – Access Control List) is a security feature used to control who can access specific data or perform certain actions in a system. It is widely used in platforms like ServiceNow to protect sensitive information.

1. What is Access Control?

Access Control defines permissions for users based on their roles. It determines whether a user can read, write, create, or delete records in a table.

2. Purpose of Access Control

- Protect sensitive data
- Restrict unauthorized access
- Ensure users perform only allowed actions
- Improve system security

3. Example in Smart Library System

- In a Smart Library Request Workflow:
- Students can only create and view their book requests.
- Librarians can approve or reject requests.
- Admins can manage all records and users.

4. Steps to Create Access Control

1. Go to System Security → Access Control (ACL)
2. Click New
3. Select the table and operation (read, write, create, delete)
4. Assign the required role
5. Save the Access Control rule

5. Benefits

Enhances data security

Controls user permissions

Prevents unauthorized modifications

Maintains system integrity

- Role creation
- Table Creation
- UI Policy
- Client Script
- Flow Designer
- Access Control.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_book	Allow If	delete	record	true	admin	2026-03-06 19:52:53
u_book	Allow If	create	record	true	admin	2026-03-06 19:52:53
u_book	Allow If	delete	record	true	admin	2026-03-06 20:59:27
u_book	Allow If	read	record	true	admin	2026-03-06 19:52:53
u_book	Allow If	write	record	true	admin	2026-03-06 19:52:53
u_book	Allow If	read	record	true	admin	2026-03-06 20:56:37
u_book	Allow If	write	record	true	admin	2026-03-06 20:58:05

- This image displays the fields that become read-only after approval.
- Fields like requested by and book details cannot be edited.
- This prevents accidental or unauthorized changes.
- It ensures the request information remains accurate.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_borrow_request	Allow If	write	record	true	admin	2026-03-06 19:58:33
u_borrow_request	Allow If	delete	record	true	admin	2026-03-06 19:58:33
u_borrow_request	Allow If	create	record	true	admin	2026-03-06 19:58:33
u_borrow_request	Allow If	read	record	true	admin	2026-03-06 19:58:33
u_borrow_request	Allow If	create	record	true	admin	2026-03-06 21:01:01
u_borrow_request	Allow If	delete	record	true	admin	2026-03-06 21:04:59
u_borrow_request	Allow If	write	record	true	admin	2026-03-06 21:03:26
u_borrow_request	Allow If	read	record	true	admin	2026-03-06 21:02:26

- This image shows the Access Control List (ACL) configuration for the Book table.
- ACL rules control permissions such as read, write, create, and delete.
- These permissions ensure only authorized users can access book records.
- This improves system security.

ACTIVITY 6 - REPORT

Report is a feature used to display system data in a visual or structured format so users can easily analyse information. In platforms like ServiceNow, reports help monitor records, activities, and system performance.

1. What is a Report?

A report collects data from tables and presents it in formats such as tables, charts, graphs, or lists. It helps users understand the status and progress of different processes.

2. Purpose of Reports

- Analyse system data
- Track requests and activities
- Monitor workflow performance
- Support decision making

3. Example in Smart Library System

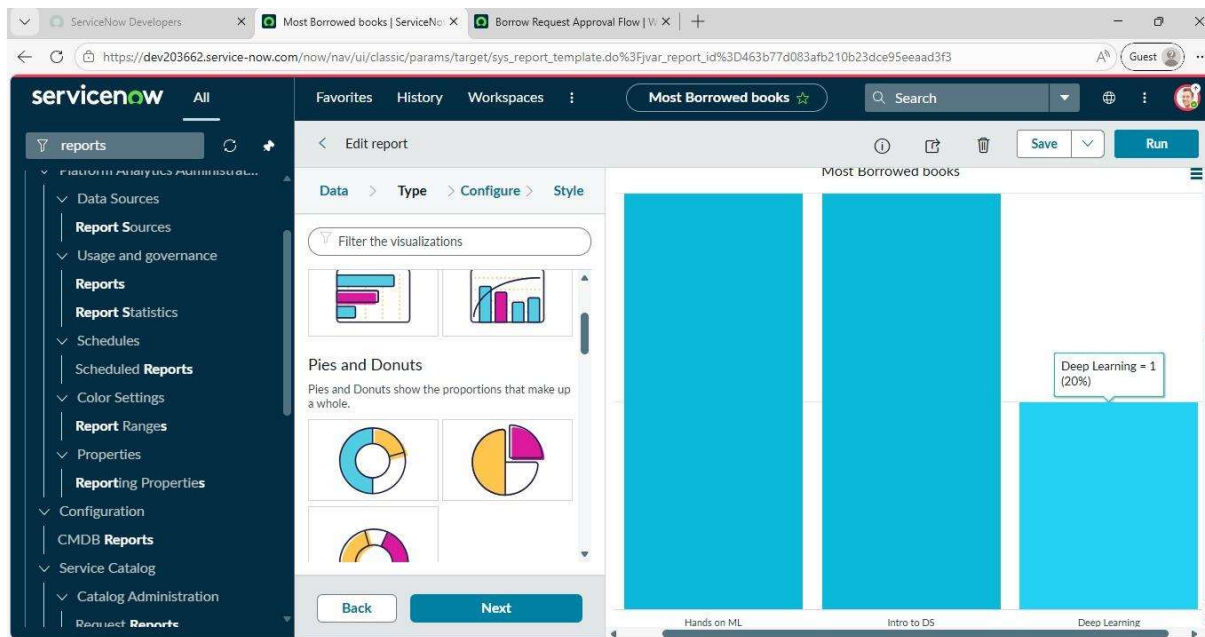
- In a Smart Library Request Workflow, reports can show:
- Total book requests submitted
- Number of approved and rejected requests
- Pending requests waiting for librarian approval
- Most requested books

4. Steps to Create a Report

1. Navigate to Reports → Create New
2. Select the table containing the data
3. Choose the report type (bar chart, pie chart, list, etc.)
4. Apply filters or conditions
5. Save and generate the report

5. Benefits

- Provides clear data visualization
- Helps track system activities
- Improves decision making
- Makes data easy to understand



- This image shows the report created in ServiceNow.
- The report displays information about the most borrowed books.
- Charts or graphs are used to visualize the data.
- Reports help administrators analyse library usage and make better decisions.

ServiceNow workflows automate the approval process. Requests are automatically routed to the appropriate librarian for review.

Reporting Module

This module generates reports and dashboards showing statistics such as most borrowed books, pending requests, and overdue returns.

7. Implementation

The implementation of the Smart Library Request Workflow involves several steps within the ServiceNow platform.

- Creating database tables to store information about books, students, and requests.
 - Designing forms for users to submit book requests.
 - Developing workflows to automate approval processes.
 - Configuring role-based access control to ensure security.
 - Implementing notifications for request updates.
- Through these steps, the system becomes capable of handling library operations in a structured and automated manner.

8. Testing and Results

Testing is an important stage in the development process. The system was tested to ensure that all modules function correctly and efficiently.

Different types of testing performed include:

- Functional Testing – Ensures each module performs its intended function.
- Workflow Testing – Verifies the automation of request approvals.
- User Interface Testing – Ensures forms and dashboards work properly.
- Integration Testing – Confirms communication between modules.

The results show that the system successfully automates the book request process and improves operational efficiency.

9. Advantages of the System

The Smart Library Request Workflow provides several benefits:

- Reduces manual work and paperwork.
- Improves efficiency in managing book requests.

- Provides real-time book availability information.
- Enhances transparency in the approval process.
- Enables better data analysis through reports.

10. Future Scope

The system can be further enhanced with additional features such as:

- Mobile application integration.
- Barcode or QR code scanning for book tracking.
- Integration with digital libraries and e-book platforms.
- AI-based recommendation systems for books.
- Automated reminders for overdue returns.

11. Conclusion

The Smart Library Request Workflow demonstrates how digital transformation can improve traditional library management processes. By implementing automated workflows through ServiceNow, the system simplifies book requests, approvals, and record management.

The project successfully reduces manual effort and enhances transparency in library operations. With further enhancements, the system can evolve into a fully integrated smart library platform.

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3. CRM Application for TheChaiShop&Co Project Documentation.
4. Research papers on Digital Library Management Systems.